An Endmember-Oriented Transformer Network for Bundle-Based Hyperspectral Unmixing

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***Abstract*— In recent years, researchers have focused on mitigating the impact of spectral variability (SV) on unmixing performance, leading to the development of various deep-learning-based unmixing networks. Currently, most unmixing networks that account for SV mainly rely on probabilistic generative models, which lacks specific constraints for SV and suffers from the instability of solutions generated by probabilistic models. This results in the generated endmembers or SV components lacking clear physical meaning. To avoid the problems above, we propose an endmember-oriented Transformer network (EOT-Net) that leverages the advantages of endmember bundles to introduce variability while providing stable endmember results with clear physical meaning. We design an endmember-oriented Transformer (EOT) to capture endmember-specific features through directional subspace projection and a low-redundancy attention (LRA) mechanism. Subsequently, the proposed network is divided into two branches: endmember generation and abundance estimation, to process endmember-specific features. In the endmember generation branch, endmember-specific features are transformed into intraclass weights that are used to combine signatures within the bundles, and a set of endmembers is generated for each pixel. In the abundance estimation branch, endmember-specific features are integrated using a heterogeneous information fusion (HIF) module that leverages the spatial distribution heterogeneity of the endmembers, ultimately producing the abundance results. We applied the proposed algorithm to both synthetic and real datasets, and the experimental results demonstrated the model's superiority.**

***Index Terms*— Autoencoder (AE) network, endmember bundle, hyperspectral unmixing (HU), spectral variability (SV), Transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) extends traditional 2-D images by enriching the spectral dimension, with numerous bands providing rich spectral information. The broadband range and high spectral resolution of hyperspectral imaging enable it to reflect distinct material characteristics. And a series of hyperspectral-based research and applications have been conducted, such as hyperspectral classification, change detection, and semantic segmentation. Since hyperspectral imaging is mostly based on airborne equipment, its spatial resolution is limited. As a result, a single pixel in an HSI often contains multiple substances, leading to the presence of mixed pixels. Hyperspectral unmixing (HU) technology has been proposed to explore the materials present in mixed pixels and their corresponding proportions. It is a technique that can extract the material signatures present in mixed pixels and their corresponding proportions, known as endmembers and abundances in HU.

To accurately extract endmembers and estimate abundances, understanding the mixing effects between spectra is essential. The linear mixing model (LMM) [1] is a physically meaningful and widely adopted approach. The LMM assumes that photons are reflected by only one material before entering the imaging device and do not interact with other photons. Therefore, the obtained spectra are a linear combination of endmembers, with abundances representing the corresponding mixing proportions. And a series of algorithms based on the LMM have been studied, including geometry-based algorithms [2], [3], [4], [5] and nonnegative matrix factorization (NMF)-based algorithms [6], [7], [8]. However, the assumption of linear mixing may introduce errors. Therefore, nonlinear mixing models (NLMMs) [9], [10], [11], [12], [13] have been proposed, which consider the nonlinear effects between photons. Recently, researchers are increasingly focusing on the study of the extended multilinear mixing model (EMLM). BNLSU [14] and EMLM-Net [15] approach the EMLM from the perspectives of band-wise nonlinear modeling and neural networks, respectively, further improving unmixing accuracy. NLMMs improve unmixing accuracy by modeling nonlinear effects, but they also increase the complexity of the models.

Spectral variability (SV) is a factor affecting unmixing accuracy, leading to the phenomenon of the same material having different spectra. Typically, SV is caused by changes in illumination, atmospheric conditions, or internal factors of the material. To mitigate the impact of SV, a series of unmixing algorithms have been proposed. These can be mainly categorized into: bundle-based algorithms [16], [17], [18], model-based algorithms [19], [20], [21], [22], and deep-learning-based algorithms [23], [24], [25].

Bundle-based algorithms typically extract multiple sets of endmembers from image subsets and use multiple signatures to represent one endmember [26], [27], [28]. Many algorithms have been proposed to identify representative endmember bundles. GMBUA [29] constructed endmember bundles using a group sparse-inducing mixed norm constraint. TSEA [30] employed the evolutionary algorithm within the clustered subspace to search for and construct endmember bundles, thus

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reducing computational complexity. DOEBE [17] introduced the concept of dynamic optimization, iteratively constructing endmember bundles based on reconstruction errors.

Model-based algorithms typically improve the LMM by addressing the impact of SV. The normal compositional model (NCM) [31] models the distribution of endmembers as a Gaussian distribution, the perturbed LMM (PLMM) [32] introduces perturbation terms as additional endmembers to fit irregular disturbances, and the augmented LMM (ALMM) [33] considers both regular and irregular disturbances and models them accordingly.

In recent years, deep learning methods have provided effective solutions for unmixing, thanks to their powerful fitting capabilities and outstanding performance. Specifically, the AE structure with self-supervised characteristics has been widely used in HU [34]. DeepGUn [23] learned the latent variability manifold of endmembers and estimates endmembers based on a generative model. PGMSU [35] mitigated the influence of SV by using a variational AE to generate a set of endmembers for each pixel. DGMSSU [36] combined spatial information to generate endmembers and introduced a discriminator to improve the quality of endmembers. To fully utilize the 3-D structure of HSIs, 3D-CNN/var [37] captured spatial–spectral information using 3-D convolutional neural networks and introduced variational inference based on the PLMM to generate endmembers under the influence of SV. Zhao et al. [38] further proposed modeling for additive perturbations. SPLMM [39] designed a multibranch structure to simultaneously model the scaling and perturbation terms. MAAE [40] enhanced feature extraction through spatial and spectral attention and introduced a learnable scaling term for endmembers. ULA-Net [41] proposed a central pixel-directed attention mechanism and incorporated a scaling decoder based on the extended LMM [21]. FSVA-Net [42] leveraged the feedback mechanism and an SV attention mechanism to separate and model SV from the data. The above algorithms based on generative networks and modified LMMs may generate meaningless endmembers. Rev-net [43] constructed a flow-based structure that explicitly models the endmembers affected by SV using mathematical expressions, making the network training more stable.

A. Motivation

Due to the unstable performance of probabilistic generative models, they may produce endmembers with unclear physical meanings. And algorithms based on modified LMMs lack effective constraints for addressing the scaling and perturbation effects caused by SV. The endmember bundle algorithm considers endmember variability while preserving explicit physical meanings, providing an effective approach for addressing unmixing under the influence of SV. Currently, endmember bundles play a positive role in stabilizing the training of neural networks. TANet [44] and EGU-Net [45] utilized layers specifically designed for extracting endmember bundle features, which are then used to guide the training of the unmixing branch. DGMSSU [36] utilized a discriminator and endmember bundles to promote the generation of meaningful endmembers by the model. However, limited research has explored leveraging the advantages of endmember bundles in unmixing networks for endmember generation. Therefore, we aim to design an endmember generator based on endmember bundles.

Endmember bundles utilize intraclass weights specific to each endmember class to combine signatures within the bundles, providing an effective method for endmember generation. However, the quality of the generated endmembers is influenced by the intraclass weights. Accurate intraclass weights must be derived from discriminative features of the input HSI, making efficient feature extraction algorithms crucial. Recently, Transformers [46], [47], [48] have demonstrated outstanding performance in unmixing tasks due to their ability to capture global features. DeepTrans [49] was the first to introduce Transformers into unmixing tasks, showing the positive impact of global information on unmixing. UnDAT [50] and UST-Net [51] used window-based Transformers to fully explore material distribution characteristics. The Transformer offers an effective approach for globally extracting endmember-specific features. However, many Transformer modules rely entirely on network iterations to achieve specific functions, lacking prior information to guide the attention mechanism toward valuable features. In addition, the multihead attention mechanism can lead to information redundancy. To address these issues, we propose a Transformer tailored for extracting features specific to each endmember class and introduce a novel attention module to reduce information redundancy.

In unmixing networks, abundance is typically derived from the overall features of the image, which allows for the integration of information from multiple endmembers and generates results that satisfy the physical constraints on abundance. However, the proposed network extracts endmember-specific features for each endmember class. Therefore, it is necessary to develop a feature fusion method to aggregate information from different endmember classes based on the distribution characteristics of them. Considering the complementary nature of endmembers' spatial distributions, we develop a fusion module leveraging this complementarity to provide reliable features for abundance estimation.

In summary, we aim to incorporate endmember bundles into the AE to account for SV. To accurately derive intraclass weights for each endmember bundle and generate precise endmembers, we design a Transformer-based endmember-specific feature extraction module. Since abundance estimation requires integrating information from all endmembers, we developed a feature fusion module to aggregate features from various endmembers effectively.

B. Contributions

Considering the above issues and insights, we propose the endmember-oriented Transformer network (EOT-Net). The contributions of this article are summarized as follows.

1) In this article, we propose a bundle-based unmixing network called EOT-Net to effectively addresses endmember variability while providing physically meaningful endmember results. In this work, a bundle-based

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endmember generator that leverages the diverse endmember signatures within the bundle and combines intraclass weights to generate a set of endmembers for each pixel.

2) To fully explore the characteristics of specific endmember, we introduce the concept of subspace projection and designed the endmember-oriented Transformer (EOT) module. In addition, to ensure the attention mechanism focuses on rich and discriminative information, we implement a low-redundancy attention (LRA) mechanism based on the mixture of experts (MOEs).
3) A heterogeneous information fusion (HIF) module is proposed to integrate features belonging to different endmembers. This module leverages the heterogeneity of the spatial distribution of endmembers and introduces a complementary feature fusion and channel interaction strategy, providing valuable features for abundance estimation.

The rest of this article is organized as follows. Section II reviews related works, covering both endmember bundles and the MOEs. In Section III, we thoroughly explain the architecture of our proposed EOT-Net, which includes the EOT module, the HIF module, and a bundle-based endmember generator. Section IV presents experimental results and compares the performance of our model with other state-of-the-art (SOTA) algorithms. Finally, Section V provides the conclusion of the article.

II. Related Works

A. LMM and Endmember Bundle

1) Linear Mixing Model: The LMM [1] is the most commonly considered approach in HU. This model assumes that mixed spectra result from a linear combination of endmembers, with the corresponding mixing coefficients representing the abundances. We use N to represent the number of pixels in the image and L to represent the number of bands. The original hyperspectral data can thus be denoted as $\mathbf{Y} \in \mathbb{R}^{L\times N}$. We can formulate the LMM in matrix form as follows:

$$\begin{aligned} \mathbf{Y} &= \mathbf{M}\mathbf{A} + \mathbf{R} \\ \text{s.t. } \mathbf{1}_P^{\mathrm{T}}\mathbf{A} &= \mathbf{1}_N^{\mathrm{T}}, \mathbf{A} \geq 0 \end{aligned} \tag{1}$$

where $\mathbf{M} \in \mathbb{R}^{L\times P}$ is the endmember matrix and $\mathbf{A} \in \mathbb{R}^{P\times N}$ denotes the abundance matrix. P indicates the number of endmembers. And $\mathbf{R} \in \mathbb{R}^{L\times N}$ represents the noise matrix. In addition, considering the constraints of real-world scenarios and the physical significance of abundance, the abundance must satisfy two additional constraints: the abundance sum-to-one constraint (ASC) and the abundance nonnegativity constraint (ANC).

2) Endmember Bundle: Due to the existence of SV, a single signature often fails to accurately represent the endmember in real-world scenarios. Therefore, researchers have developed LMMs based on endmember bundles, introducing variability in the endmember matrix by using multiple signatures to represent the same material. The formula can be expressed as

$$\mathbf{Y} = \left(\sum_{i=1}^{P} \mathbf{M}_i \mathbf{S}_i\right)\mathbf{A} + \mathbf{R} \tag{2}$$

where $\mathbf{M}_i \in \mathbb{R}^{L\times K}$ denotes the endmember bundle matrix of the ith material. $\mathbf{S}_i \in \mathbb{R}^{K\times N}$ is the corresponding intraweights of signatures in the bundle. And K represents the number of signatures in an endmember bundle. Similarly, the abundance should satisfy two constraints in the LMM.

B. Mixture of Experts

MOEs [52], [53] is a widely used technology in machine learning. MOE consists of multiple specialized submodules, each addressing the same task from different perspectives. This can be seen as collaborative work among multiple experts. In addition, MOE uses a weighting function $w(\cdot)$ to calculate the contribution of each expert. This design allows tasks to be delegated to more specific models, reducing overfitting and enhancing the model's generalization capability

$$\begin{aligned} y &= \sum_{i=1}^{n} w_i(x) f_i(x) \\ \sum_{i=1}^{n} w_i(x) &= 1 \end{aligned} \tag{3}$$

where $f_i(\cdot)$ denotes the ith expert in MOE and n is the number of experts in MOE. $w(x) \in \mathbb{R}^{1\times n}$ represents the weights for each expert, with the sum of these weights being one.

III. Proposed Method

This section presents an introduction to our proposed network, which consists of an EOT module, an HIF module, and a bundle-based endmember generator.

A. Network Architecture

The overall architecture of the network is shown in Fig. 1, which uses two branches to extract endmembers and estimate abundance. In this work, we introduce variability to the extracted endmembers by using endmember bundles. We designed an EOT to thoroughly explore the internal information of each material, enabling feature extraction specific to each endmember class. The extracted features are then used to generate intraclass weights for each endmember bundle, and a bundle-based endmember generator estimates endmembers for every pixel. In addition, after extracting class-specific features, we employ an HIF module to achieve spatial complementarity and interaction across channels. The abundance is then estimated based on the fused features. We will provide a detailed introduction to the proposed model in Sections III-B–III-D.

Given the original HSI $\mathbf{Y} \in \mathbb{R}^{N\times L}$, we use a linear layer to project it into a lower dimension. L represents the number of bands. And N indicates the number of pixels in the input. We can formulate this processing as

$$\mathbf{X} = \mathrm{Proj}(\mathbf{Y}) \tag{4}$$

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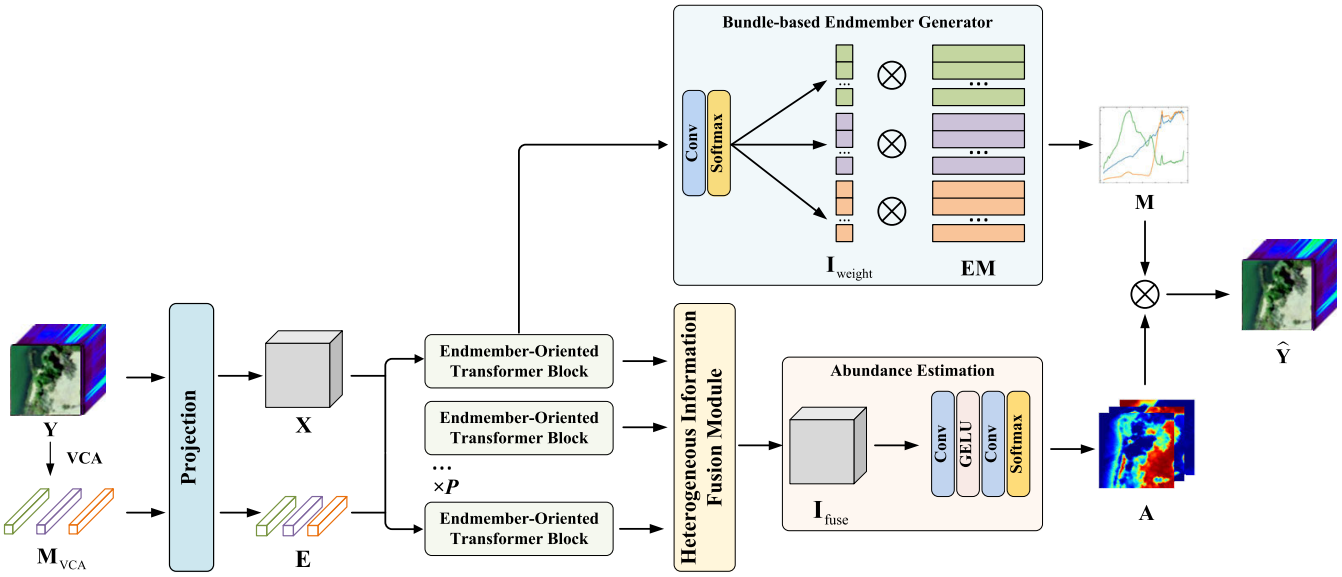


Fig. 1. Architecture of the proposed EOT-Net that leverages several parallel EOT blocks to extract features specific to each endmember. The bundle-based endmember generator transforms the intraclass features of each endmember into intrabundle weights, which weight the endmember bundles and provide a set of endmembers for each pixel. In addition, the HIF module integrates features by exploiting the spatial heterogeneity of endmember distributions, leading to the estimated abundances based on the fused features.

where $\mathbf{X} \in \mathbb{R}^{N\times D}$ represent the feature of the input after dimensionality reduction. And D is the number of channels in $\mathbf{X}$, which is chosen to be 32 in the proposed model.

To stabilize the network's training process and achieve endmember-oriented feature extraction, we utilize a classic endmember extraction algorithm (such as vertex component analysis (VCA) [4]) to obtain reference endmembers

$$\mathbf{M}_{\text{VCA}} = \text{VCA}(\mathbf{Y}) \tag{5}$$

where $\mathbf{M}_{\text{VCA}} \in \mathbb{R}^{P\times L}$ is the reference endmembers extracted by VCA. Similarly, to utilize the reference endmembers in the feature space, we project $\mathbf{M}_{\text{VCA}}$ to the same dimension as $\mathbf{X}$. It is important to note that we use a projection layer with shared weights to ensure they are projected into the same feature space. We formulate the operation as follows:

$$\mathbf{E} = \text{Proj}(\mathbf{M}_{\text{VCA}}) \tag{6}$$

where $\mathbf{E} \in \mathbb{R}^{P\times D}$ is the endmember token, which represents the feature of the reference endmembers and is used to guide the extraction of information for each endmember class.

B. Endmember-Oriented Transformer

In the proposed model, we attempt to extract useful information belonging to each endmember class, which is subsequently used in the bundle-based unmixing module. Therefore, we design an EOT module. This module, guided by endmember tokens, suppresses heterogeneous information and extracts effective information relevant to the current endmember class based on its correlations. In addition, to address redundancy issues in multihead attention, we propose an LRA mechanism and design a redundancy-reducing module based on an MOE layer.

To be more detailed, the architecture of the proposed EOT is shown in Fig. 2. Based on the number of endmembers, $\mathbf{X}$ is fed into multiple EOT blocks. First, we design an endmember-oriented projection (EOP) to suppress information from other endmembers by projecting into the feature space corresponding to the current endmember, thereby highlighting the features of the current class. Unlike the unstable projections based on fully connected layers in other neural networks, we design a directional projection layer. The projection operator $\mathbf{P}_{\text{oriented}}$ is constructed by the endmember token of the current class and can be expressed as follows:

$$\mathbf{P}_{\text{oriented}} = (\mathbf{E}_i)^{\text{T}} \cdot (\mathbf{E}_i(\mathbf{E}_i)^{\text{T}})^{-1} \cdot \mathbf{E}_i \tag{7}$$

where $\mathbf{E}_i$ is the ith vector in $\mathbf{E}$. And $(\cdot)^{\text{T}}$ denotes the transpose of a matrix. $(\cdot)^{-1}$ is the inverse of a matrix. $\mathbf{P}_{\text{oriented}}$ represents the algorithm for projecting onto the subspace of $\mathbf{E}_i$, derived from the least squares method. $\mathbf{E}_i$ can be considered as the basis vectors of the subspace.

To enhance the model's stability and generalization, we apply layer normalization (LN) to the projected data

$$[\mathbf{E}_i', \mathbf{X}'] = \text{LN}(\mathbf{P}_{\text{oriented}} \cdot [\mathbf{E}_i, \mathbf{X}]) \tag{8}$$

where [ ] represent the concatenation operation.

Then, the endmember and data feature are sent to a low-redundancy attention mechanism for endmember-oriented feature extraction. In the proposed LRA, we project the endmember token as the query vector and use the data feature as the key and value matrices. This design guides the attention mechanism to focus more on the information related to the current class while suppressing information corresponding to other endmembers

$$\mathbf{Q} = \mathbf{E}_i'\mathbf{W}_{\text{q}}, \quad \mathbf{K} = \mathbf{X}'\mathbf{W}_{\text{k}}, \quad \mathbf{V} = \mathbf{X}'\mathbf{W}_{\text{v}} \tag{9}$$

where $\mathbf{W}_{\text{q}}, \mathbf{W}_{\text{k}}, \mathbf{W}_{\text{v}} \in \mathbb{R}^{D\times(D\cdot h)}$ are the projection matrices, h is the number of attention heads (h is exemplified as 4 in Fig. 2), and D is the number of channels for every attention

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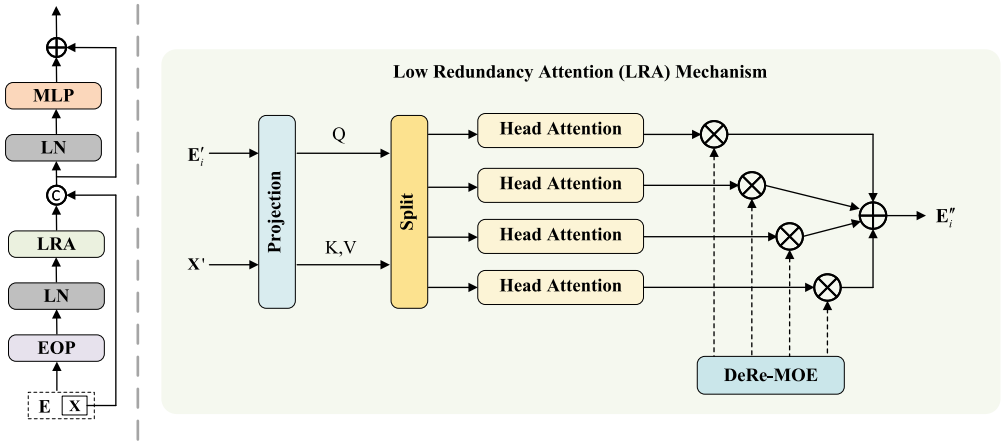


Fig. 2. Detailed structure of the proposed EOT. This module uses an EOP to map the data into feature subspaces corresponding to each endmember. Unlike traditional attention mechanisms, it introduces an LRA mechanism, employing an MOEs-based de-redundancy module.

head. Similar to other multihead attention algorithms, we split the above matrices into multiple attention heads to enable information extraction across multiple feature spaces

$$\text{head}_1, \ldots, \text{head}_h = \text{Split}([\mathbf{Q}, \mathbf{K}, \mathbf{V}]) \tag{10}$$

where $\text{head}_i = [\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i]$ denotes the ith attention head, and it is composed of $\mathbf{Q}_i \in \mathbb{R}^{1\times D}$, $\mathbf{K}_i, \mathbf{V}_i \in \mathbb{R}^{N\times D}$ for each attention head

$$\text{head}_i' = \text{Softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^{\mathrm{T}}}{\sqrt{D}}\right)\mathbf{V}_i \tag{11}$$

where $\text{head}_i' \in \mathbb{R}^{1\times D}$ is the ith attention head after attention weighting. Based on the calculation, this mechanism applies attention weighting to the data according to the correlation between the global data and the current endmember token.

Different from the standard multihead attention mechanism that concatenates multiple attention heads as the final result, we design a de-redundancy-MOE (DeRe-MOE) based on an MOEs system to fuse useful information and avoid redundancy in the attention heads. First, we perform a similarity assessment for each attention head's result. Considering that each attention head should extract information from different feature spaces to maximize the entropy of the extracted features, we use spectral information divergence (SID) for similarity evaluation from an information theory perspective. The formula is given as follows ($\mathbf{p}$ and $\mathbf{q}$ are vectors with D channels):

$$\text{SID}(\mathbf{p}, \mathbf{q}) = \sum_{i=1}^{D}\left(p_i \log\left(\frac{p_i}{q_i}\right) + q_i \log\left(\frac{q_i}{p_i}\right)\right). \tag{12}$$

Thus, a similarity matrix $\text{Sim} \in \mathbb{R}^{h\times h}$ is obtained, where each element of the matrix represents the SID between the corresponding attention heads. The proposed DeRe-MOE converts the similarity matrix into weights for each attention head, aiming to filter useful information and fully optimize the model's performance. Specifically, the formula is given as follows:

$$g = 1 - \text{Softmax}(\text{FC}(\text{Sim})) \tag{13}$$

$$\text{Softmax}(w_i) = \frac{\exp(w_i)}{\sum_{j=1}^{h}\exp(w_j)} \tag{14}$$

where $g \in \mathbb{R}^{h\times 1}$ represents the gating weights of the heads. In DeRe-MOE, a fully connected layer and the nonlinear function Softmax are utilized to transform the similarity matrix Sim into gating weights. We aim to preserve information from different feature spaces, ensuring that the differences between attention heads are salient. Therefore, we use 1 minus the weights derived from the similarity matrix as the final w. The fused attention-weighted result $\mathbf{E}_i'' \in \mathbb{R}^{D\times 1}$ is obtained based on the following formula:

$$\mathbf{E}_i'' = \sum_{i=1}^{h} g_i \cdot \text{head}_i. \tag{15}$$

Furthermore, to achieve feature interaction between channels and enhance the model's representational capacity, a multilayer perceptron (MLP) is utilized

$$\left[\mathbf{E}_i''', \mathrm{I}\right] = \left[\mathbf{E}_i'', \mathbf{X}\right] + \text{MLP}\big(\text{LN}\big(\left[\mathbf{E}_i'', \mathbf{X}\right]\big)\big). \tag{16}$$

We use the endmember-oriented feature $\mathbf{I} \in \mathbb{R}^{D\times N}$ extracted by the module as the output of the EOT block. Therefore, for P parallel EOT blocks, the module's output can be represented as

$$\mathbf{I}_i = \text{EOT}(\mathbf{E}_i, \mathbf{X}) \tag{17}$$

where $\mathbf{I}_i$ denotes the ith endmember-oriented feature for the ith endmember. Therefore, the module outputs P features: $\mathbf{I}_1, \ldots, \mathbf{I}_P$.

C. HIF Module

Due to the distinct spatial distributions of different endmembers, there is evident heterogeneity among the extracted endmember-oriented features. Since the estimation of abundance requires considering the features of various endmembers, we devise an HIF module to aggregate features of multiple endmembers. The proposed HIF module is shown in Fig. 3. We perform feature fusion in both the spatial and channel dimensions. To account for the complementary characteristics of material distribution in space, we devise a

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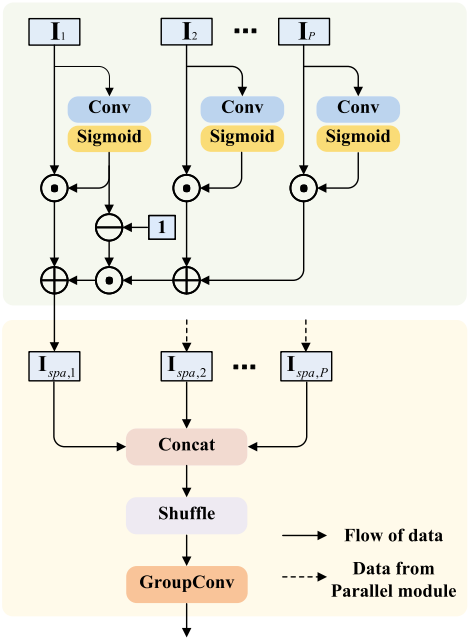


Fig. 3. HIF module using $\mathbf{I}_1$ as an example. In the figure, solid lines indicate the flow of data, while dashed lines represent the outputs from other parallel modules for spatial fusion.

spatial complementary fusion part. In addition, considering that each channel of the features represents a different feature representation, we also design a channel interaction fusion part.

Spatial complementary fusion primarily focuses on extracting valuable information from other endmember-oriented features that contribute to the current endmember. For the sake of illustration, let's take the spatial fusion of the ith feature $\mathbf{I}_i$ as an example. First, $\mathbf{I}_i$ is reshaped into a 3-D tensor to better represent its spatial and channel features, that is $\mathbf{I}_i \in \mathbb{R}^{N\times D} \rightarrow \mathbb{R}^{H\times W\times D}$. Then, we employ a 1×1 convolution and the Sigmoid function to process the feature map and obtain a mask. When enhancing the ith feature $\mathbf{I}_i$, the mask can be divided into two types. The first type is the self-based mask $\boldsymbol{m}_i$, which represents the valuable parts within the current feature $\mathbf{I}_i$. The second type of mask $\boldsymbol{m}_{j\rightarrow i}$ represents the information from the $\mathbf{I}_j$ required for enhancing the $\mathbf{I}_i$. The mask generation process can be represented by the following equations:

$$\boldsymbol{m}_i = \text{Sigmoid}(\text{Conv}(\mathbf{I}_i)) \tag{18}$$

$$\boldsymbol{m}_{j\rightarrow i} = \text{Sigmoid}(\text{Conv}(\mathbf{I}_j)) \tag{19}$$

where $\boldsymbol{m}_i \in \mathbb{R}^{H\times W}$ and $\boldsymbol{m}_{j\rightarrow i} \in \mathbb{R}^{H\times W}$ are the two types of masks, both having the same spatial dimension as the input.

To enhance the regions where the current feature does not contribute useful information, we introduce spatial complementary fusion, utilizing the complement of $\boldsymbol{m}_i$. This approach enhances those regions by leveraging information from other features

$$\mathbf{I}_{\text{spa},i} = \mathbf{I}_i + \boldsymbol{m}_i \odot \mathbf{I}_i + (1-\boldsymbol{m}_i) \odot \sum_{j\neq i}^{P} (\boldsymbol{m}_{j\rightarrow i} \odot \mathbf{I}_j) \tag{20}$$

where $\mathbf{I}_{\text{spa},i} \in \mathbb{R}^{H\times W\times D}$ is the spatially fused feature of the ith endmember, and $\odot$ denotes the Hadamard product used for element-wise multiplication. After applying the aforementioned operations to each endmember-oriented feature, the spatial complementary fusion module generates P spatially fused features $\mathbf{I}_{\text{spa},1}, \ldots, \mathbf{I}_{\text{spa},P}$.

To achieve channel interactions, we concatenate the features along the channel dimension

$$\mathbf{I}_{\text{spa}} = \text{Concat}(\mathbf{I}_{\text{spa},1}, \ldots, \mathbf{I}_{\text{spa},P}) \tag{21}$$

where $\mathbf{I}_{\text{spa}} \in \mathbb{R}^{H\times W\times (P\cdot D)}$ represents the feature after concatenating. Since each feature's different channels represent unique feature information, we utilize channel shuffling to prevent the loss of valuable information during fusion. By employing group convolution, we achieve information fusion across channels, resulting in the output of the fused features

$$\mathbf{I}_{\text{fuse}} = \text{GConv}(\text{CS}(\mathbf{I}_{\text{spa}})) \tag{22}$$

where $\mathbf{I}_{\text{fuse}} \in \mathbb{R}^{H\times W\times D}$ denotes the feature after HIF, $\text{CS}(\cdot)$ refers to channel shuffle operation, and $\text{GConv}(\cdot)$ is the group convolution step.

The module decodes the fusion features, which combine information from multiple endmembers, into estimated abundance. The specific structure is depicted in Fig. 1 for clarity and visualization. The formula is expressed as

$$\mathbf{A} = \text{Softmax}(\text{Conv}(\text{GELU}(\text{Conv}(\mathbf{I}_{\text{fuse}})))) \tag{23}$$

where $\mathbf{A} \in \mathbb{R}^{H\times W\times P}$ denotes the estimated abundance tensor, Softmax is used to impose the sum to one constraint, GELU is a nonlinear function, and convolutions are is utilized for aggregating spatial information and reducing dimensionality.

D. Bundle-Based Endmember Generator

To incorporate the variability within endmembers and ensure their physical interpretation, we propose a bundle-based endmember generator. Its structure is illustrated in the upper branch of Fig. 1. Specifically, for each endmember, we design a fully connected layer with a size of $K \times L$ to represent an endmember bundle. In other words, K signatures are used to represent one endmember. Therefore, for the HSI with P endmembers, the module requires P fully connected layers $\mathbf{EM}_1, \ldots, \mathbf{EM}_P \in \mathbb{R}^{K\times L}$. Furthermore, to stabilize the network training and ensure that the generated endmembers from the bundle-based endmember generator have more meaningful physical interpretations, we initialize the weights of the fully connected layers using endmember bundles preextracted by the bundle extraction algorithm. The algorithm partitions the original data into multiple superpixels and extracts endmembers within each superpixel based on the VCA algorithm. The endmembers within each superpixel are then clustered, and K discriminative signatures are selected as the endmember bundles.

The intraclass weights of the endmember bundles are derived based on endmember-oriented features. Since the features associated with each endmember contain material information from various spatial locations, we employ convolution and the Softmax function to generate the intraclass weights of the endmember bundles. These weights can be seen as coefficients corresponding to the endmember bundle

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dictionary, which are used to generate the endmembers at all positions in the image

$$\mathbf{I}_{\text{weight},i} = \text{Softmax}(\text{Conv}(\mathbf{I}_i)) \tag{24}$$

where $\mathbf{I}_{\text{weight},i} \in \mathbb{R}^{H\times W\times K}$ represents the coefficients corresponding to the ith endmember bundle at each position within the image. Therefore, the endmembers generated by the model can be obtained by weighting endmember bundles accordingly

$$\mathbf{M}_i = \mathbf{I}_{\text{weight},i} \cdot \mathbf{EM}_i \tag{25}$$

$$\mathbf{M} = \text{Concat}(\mathbf{M}_1, \ldots, \mathbf{M}_P) \tag{26}$$

where $\mathbf{M}_i \in \mathbb{R}^{H\times W\times L}$ represents the spectra corresponding to the ith endmember at each position. Then, we concatenate the endmember matrices of P categories to obtain the generated endmembers $\mathbf{M} \in \mathbb{R}^{H\times W\times L\times P}$ by the model. The proposed model takes into account the SV by generating a set of endmembers for each position, thereby reducing the impact of SV on abundance estimation.

E. Loss Function

An appropriate loss function guides the convergence of the model. To facilitate the computation of the reconstruction loss, we first obtain the reconstructed image $\widehat{\mathbf{Y}} \in \mathbb{R}^{H\times W\times L}$ based on the generated endmembers and estimated abundances. For ease of computation, the reconstructed image is reshaped into matrix form $\widehat{\mathbf{Y}} \in \mathbb{R}^{H\times W\times L} \rightarrow \mathbb{R}^{N\times L}$.

To thoroughly evaluate the quality of the reconstructed images produced by the model, we select the mean square error (mse) and spectral angle distance (SAD) as loss functions, which assess the loss from both numerical and spectral shape perspectives

$$\mathcal{L}_{\text{mse}} = \frac{1}{N}\sum_{i=1}^{N}\left(\widehat{\mathbf{Y}}_i - \mathbf{Y}_i\right)^2 \tag{27}$$

$$\mathcal{L}_{\text{SAD}} = \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\left\langle \mathbf{Y}_i, \widehat{\mathbf{Y}}_i\right\rangle}{\|\mathbf{Y}_i\|_2\left\|\widehat{\mathbf{Y}}_i\right\|_2}\right). \tag{28}$$

Due to the varying characteristics of different datasets, we introduce hyperparameters to weight the two loss functions, combining them into an overall loss function

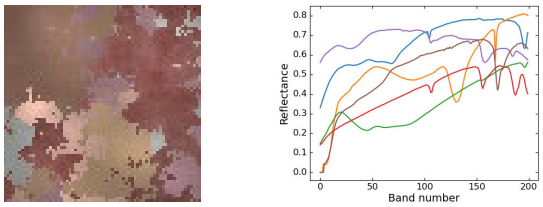
$$\mathcal{L}_{\text{all}} = \alpha\mathcal{L}_{\text{mse}} + \beta\mathcal{L}_{\text{SAD}}. \tag{29}$$

IV. Experiments

To demonstrate the performance of the proposed model, we conduct experiments on multiple datasets and compare the results with various SOTA methods. In addition, ablation experiments are performed to validate the effectiveness of each module.

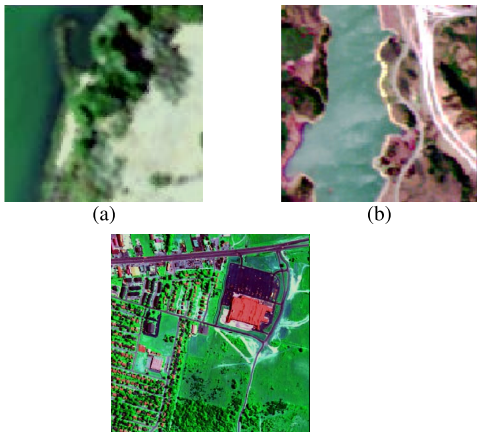
A. Hyperspectral Dataset

1) Synthetic Dataset: The synthetic dataset has spatial dimensions of 70 × 70 pixels and contains 200 spectral bands. The endmember spectra are sampled from the United States Geological Survey (USGS) mineral spectral library.



(a) (b)

Fig. 4. Synthetic dataset: (a) pseudo-color image and (b) endmembers.



(a) (b) (c)

Fig. 5. Real dataset: (a) Samson dataset, (b) Jasper Ridge dataset, and (c) Urban dataset.

A Gaussian field toolbox [54] is used to generate the abundance that satisfies the sum-to-one constraint and nonnegative constraint. To better simulate real scenarios, SV factors, and random noise are also incorporated into this dataset. For SV factors, the generated dataset primarily introduces scaling caused by illumination changes, implemented using a 2-D Gaussian distribution with a magnitude of 0.3. Random noise is simulated using Gaussian-distributed random numbers. The pseudo-color image and endmembers of the generated dataset are shown in Fig. 4(a) and (b).

2) Samson Dataset: The Samson dataset [55], shown in Fig. 5(a), is a commonly used dataset. It consists of 156 spectral bands with a spatial resolution of 95 × 95 pixels. The dataset contains three endmembers: soil, tree, and water.

3) Jasper Ridge Dataset: The Jasper Ridge dataset [55] consists of four endmembers: tree, water, dirt, and road. It contains a total of 100 × 100 pixels and 198 spectral bands. A pseudo-color image of the dataset is shown in Fig. 5(b).

4) Urban Dataset: The Urban dataset in Fig. 5(c) is composed of various urban landscapes and covers a larger area. It consists of 307 × 307 pixels and includes 162 spectral bands. This dataset consists of five endmembers: asphalt road, grass, tree, roof, and dirt, with a more complex distribution.

B. Baselines and Metrics

1) Baselines: To thoroughly demonstrate the performance of the proposed algorithm, we carefully selected seven benchmark algorithms for comparison. All experimental results are averaged over ten repeated trials. Below is a brief introduction to the comparison algorithms.

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1) *VCA-FCLS [3], [4]:* Endmember extraction and abundance estimation algorithms based on the LMM.
2) *MiSiCNet [56]:* A CNN-based unmixing network that incorporates a minimum simplex volume penalty.
3) *IMPSO [57]:* An endmember bundle extraction algorithm based on improved multiobjective particle swarm optimization.
4) *DOEBE [17]:* An endmember bundle extraction algorithm with dynamic optimization mechanism to capture the endmember variability.
5) *PGMSU [35]:* An unmixing network mitigating the impact of SV based on a generative model.
6) *PPM [58]:* A multibranch network combining the generative model and the LMM incorporating scaling factors and perturbation terms.
7) *DGMSSU [36]:* An unmixing network utilizing endmember bundles as a pr to guide the training of the Transformer-based module.

2) Metrics: In HU tasks, endmember extraction and abundance estimation are two essential steps. To quantitatively assess the model's performance, we have selected the root mse (RMSE) between estimated abundances and ground truth, as well as the SAD between endmembers and ground truth, as evaluation metrics. The RMSE of abundance (aRMSE) can be expressed as

$$\text{aRMSE} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{1}{P}\sum_{j=1}^{P}\left(a_{ji}-\widehat{a}_{ji}\right)^2} \tag{30}$$

where a_{ji} and $\widehat{a}_{ji}$ denote the true abundance vector and estimated abundance vector of the jth endmember in the ith pixel. And N and P are the number of pixels and endmembers in the dataset. The SAD of endmembers (eSAD) can be expressed as

$$\text{eSAD} = \frac{1}{P}\sum_{j=1}^{P}\arccos\left(\frac{\mathbf{m}_j^T\widehat{\mathbf{m}}_j}{\|\mathbf{m}_j\|_2\|\widehat{\mathbf{m}}_j\|_2}\right) \tag{31}$$

where $\mathbf{m}_j$ and $\widehat{\mathbf{m}}_j$ denote the true spectrum and the estimated spectrum of the jth endmember.

C. Hyperparameter Settings

In the proposed network, we introduce two hyperparameters in the loss function to adjust the contributions of different constraints. For the mse of the reconstructed image, the hyperparameter α regulates its significance, while β controls the influence of the SAD loss. Moreover, this AE framework is trained in a self-supervised manner, with the number of epochs set to 200. The network is optimized using the Adam optimizer, with a learning rate decay schedule. The number of feature channels D is set to 32, and the number h of attention heads in the attention mechanism is set to 4. The optimal parameters for each dataset are obtained through grid search, and the final values are presented in Table I. In addition, we display the parameter optimization curve for the synthetic dataset in Fig. 6. Within the reasonable parameter range shown in the figure, we determine the final hyperparameters based on the optimization process.

TABLE I
HYPERPARAMETER SETTINGS FOR DATASETS

Parameters	Synthetic	Samson	Jasper Ridge	Urban
α	0.2	0.1	0.8	0.8
β	0.6	1.0	1.0	0.4
learning rate	2e-3	2e-3	5e-3	3e-3
epoch	200	200	200	200

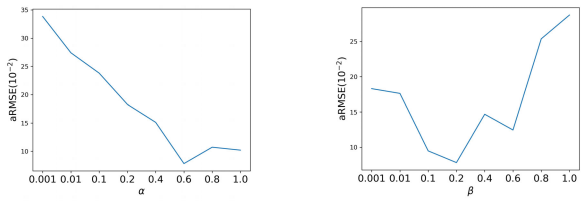


Fig. 6. RMSE of estimated abundance for different α and β.

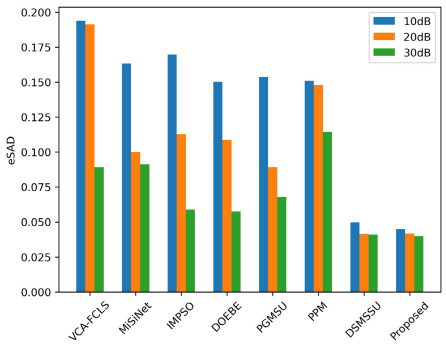


Fig. 7. Endmember extraction results presented as histograms for different algorithms across various SNRs.

D. Experiments on Synthetic and Real Datasets

In this experiment, the selected algorithms are designed with various effective endmember extraction strategies to address the variability caused by SV. IMPSO and DOEBE employ a particle swarm optimization-based endmember bundle extraction strategy to identify representative endmembers. In the unmixing network based on AEs, MiSiCNet introduces a minimum simplex constraint for endmember extraction. PGMSU and PPM utilize generative models to mitigate the impact of SV by generating a set of endmembers for each pixel. DGMSSU and the proposed algorithm incorporate endmember bundles to guide the training of the unmixing AE.

1) Synthetic Dataset: To evaluate the model's performance, we conduct experiments on synthetic datasets with varying signal-to-noise ratios (SNRs). The results of different algorithms across various SNRs are shown in Fig. 7. Since IMPSO and DOEBE are endmember bundle extraction algorithms that only provide endmember extraction results, and their abundance estimation relies on the performance of other algorithms, we focus on comparing the endmember extraction performance of each algorithm under different SNRs. In the high-SNR condition, endmember bundle-based algorithms such as IMPSO, DOEBE, DGMSSU, and the proposed method achieve high endmember extraction accuracy. As the SNR decreases, the performance of all algorithms decline. Among them, AE-based networks show better results in low-SNR conditions.

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TABLE II
ABUNDANCE RMSE RESULTS ON SAMSON DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA-FCLS	MiSiCNet	IMPSO	DOEBE	PGMSU	PPM	DGMSSU	Proposed
Soil	0.2014	0.1198	0.1560	0.1294	0.2064	0.1909	0.1607	**0.1090**
Tree	0.1210	0.1052	0.1201	**0.0619**	0.1825	0.1910	0.1524	0.0758
Water	0.2817	0.2082	0.1044	0.1475	0.2838	0.3121	0.2643	**0.0805**
Mean RMSE	0.2014	0.1444	0.1268	0.1129	0.2242	0.2313	0.1925	**0.0896**

TABLE III
ENDMEMBER SAD RESULTS ON SAMSON DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA	MiSiCNet	IMPSO	DOEBE	PGMSU	PPM	DGMSSU	Proposed
Soil	0.0259	**0.0123**	0.0963	0.0546	0.0307	0.1764	0.0196	0.0232
Tree	0.0961	**0.0462**	0.0999	0.0813	0.0560	0.0958	0.0858	0.0811
Water	0.1554	0.3734	**0.0896**	0.0923	0.2096	0.2617	0.1323	0.1169
Mean SAD	0.0925	0.1440	0.0953	0.0761	0.0987	0.1780	0.0793	**0.0737**

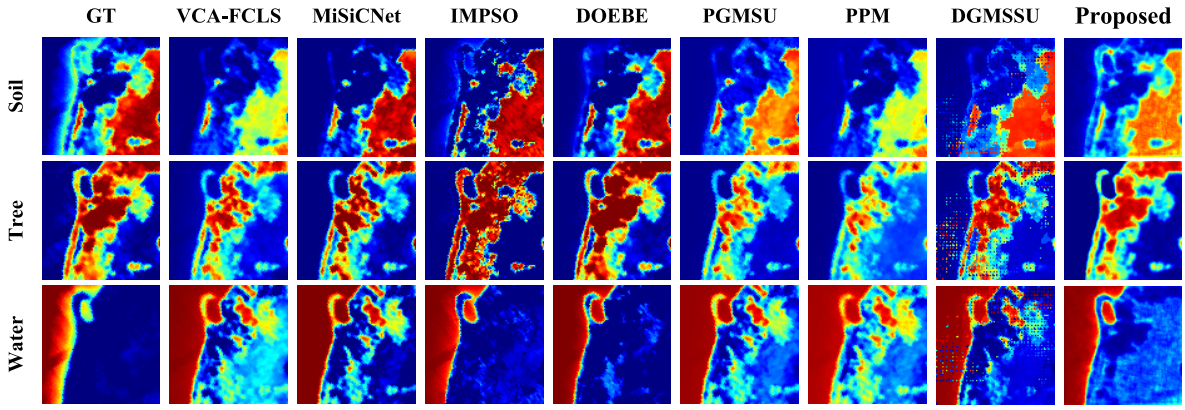


Fig. 8. Abundance map of Samson dataset.

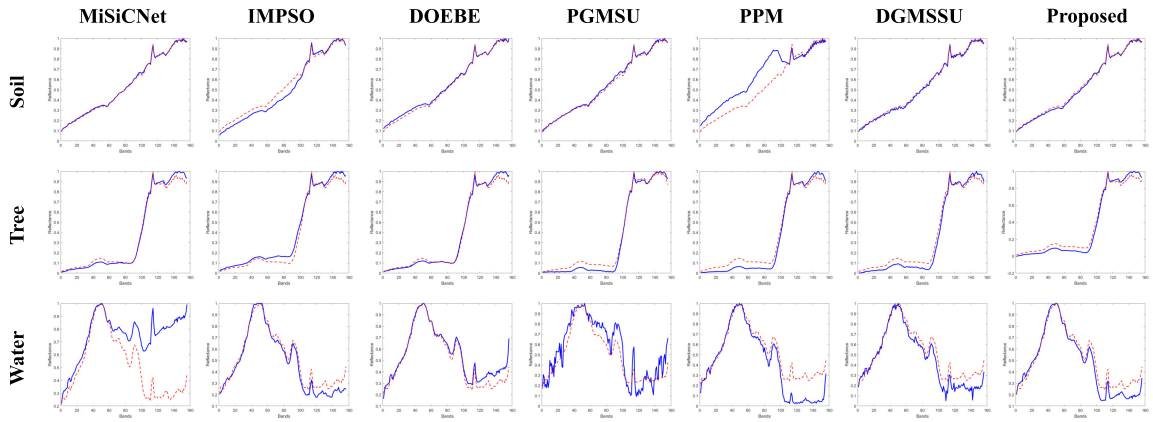


Fig. 9. Endmember curve of Samson dataset, where the red line is the true curve, and the blue line is the extraction results of different methods.

Compared to SV-based models, the endmember bundle-based AE networks demonstrate stronger robustness. The results on the synthetic dataset demonstrate that the AE combined with endmember bundles can better extract representative spectra and mitigate the impact of noise on unmixing.

2) Samson Dataset: To accurately evaluate the performance of each model, we calculate the quantitative metrics for abundance estimation and endmember extraction for each material, as shown in Tables II and III. The proposed algorithm achieves the best overall metrics in both abundance and endmember extraction. Since IMPSO and DOEBE are endmember bundle extraction algorithms, we follow the original settings in their papers and use the SUnSAL [59] algorithm for abundance estimation during the experiments. According to the results in Table II, MiSiCNet displays better performance in analyzing soil and tree. Due to the smaller number of endmembers in

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TABLE IV
ABUNDANCE RMSE RESULTS ON JASPER RIDGE DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA-FCLS	MiSiCNet	IMPSO	DOEBE	PGMSU	PPM	DGMSSU	Proposed
Tree	0.1360	0.0669	0.1183	0.0792	0.1084	0.0907	0.1093	**0.0607**
Water	0.0767	**0.0729**	0.0819	0.0778	0.1090	0.1332	0.1042	0.0793
Dirt	0.2202	0.1165	0.1434	0.1099	0.2199	0.1105	0.1154	**0.1018**
Road	0.2137	0.1435	0.1230	0.1112	0.2315	0.1274	**0.0572**	0.0742
Mean RMSE	0.1617	0.0999	0.1167	0.1018	0.1738	0.0943	0.0965	**0.0804**

TABLE V
ENDMEMBER SAD RESULTS ON JASPER RIDGE DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA	MiSiCNet	IMPSO	DOEBE	PGMSU	PPM	DGMSSU	Proposed
Tree	0.2089	**0.0664**	0.1472	0.0846	0.1230	0.1571	0.1087	0.0693
Water	0.1686	0.1692	0.1332	0.1519	0.1204	**0.0957**	0.1480	0.1380
Dirt	0.2017	0.0859	0.1496	0.1417	0.1265	0.1057	0.0397	**0.0604**
Road	0.4266	0.3772	0.0747	0.0700	0.4520	0.1592	0.0336	**0.0193**
Mean SAD	0.2515	0.1747	0.1262	0.1121	0.2055	0.1294	0.0825	**0.0717**

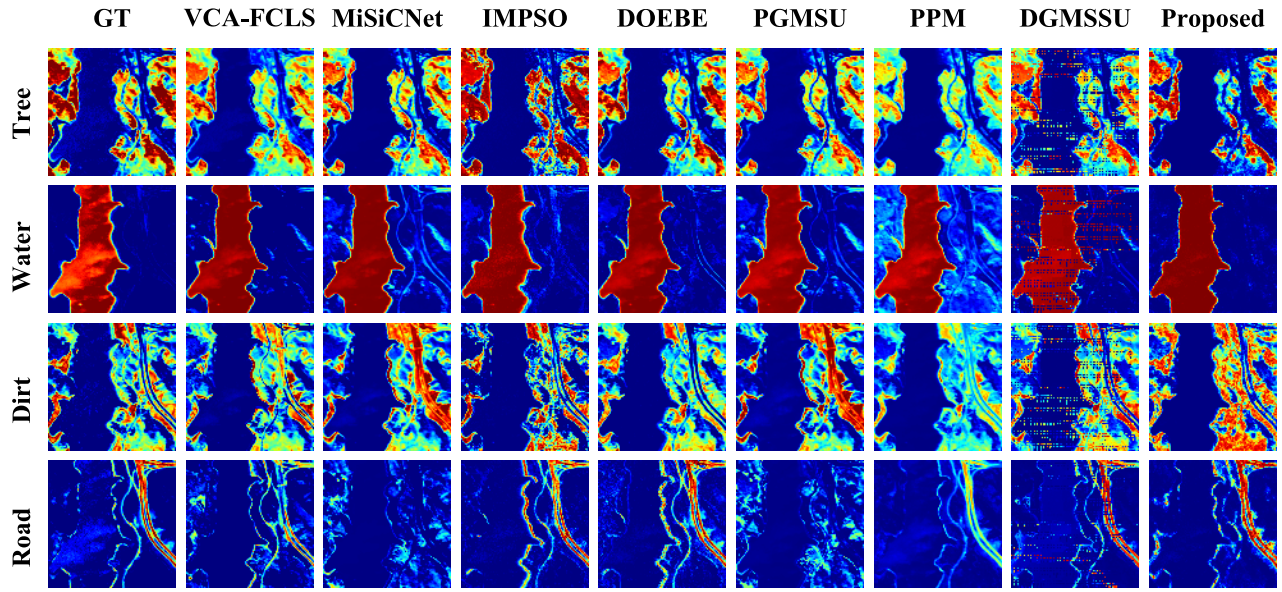


Fig. 10. Abundance map of Jasper Ridge dataset.

the Samson dataset and the relatively weak influence of SV, applying a minimum simplex constraint to the extracted endmembers maintained strong unmixing performance. However, its performance is suboptimal for the more challenging extraction of the water endmember. On the other hand, algorithms such as PGMSU and PPM, which account for SV, introduce additional parameters for SV modeling, leading to less optimal performance on simpler datasets. Unmixing algorithms based on endmember bundles, such as IMPSO, DOEBE, DGMSSU, and the proposed method, maintain stronger performance. In particular, these methods achieved higher overall accuracy in endmember extraction. It is worth noting that the proposed algorithm uses an endmember bundle-based decoder for unmixing. As a result, the quantitative metrics for endmembers are computed by averaging the SAD between all signatures in the bundle and the ground truth. Given the SV, the endmember decoder accounts for multiple spectral signatures to represent each endmember. Therefore, while the proposed algorithm may not achieve the best results for every individual endmember, its overall performance is optimal.

To visually demonstrate the differences in unmixing results across various algorithms, we present the abundance estimation and endmember extraction results in Figs. 8 and 9. Estimating the water abundance in the Samson dataset is particularly challenging. Most algorithms fail to estimate its abundance accurately, leading to lower purity on the right side of the water's abundance map. In the abundance estimation results, methods based on endmember bundles consistently show higher purity. In addition, due to the superpixel segmentation used in the DGMSSU algorithm, some regions are misclassified. The endmember spectral curves are consistent with the quantitative metrics, as most algorithms successfully

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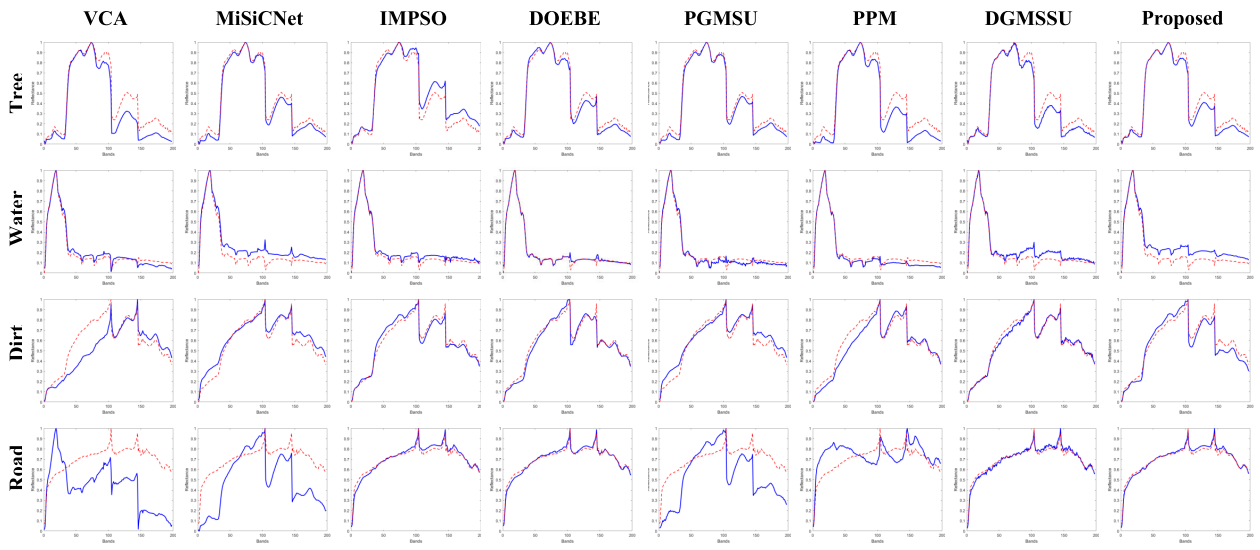


Fig. 11. Endmember curve of Jasper Ridge dataset, where the red line is the true curve, and the blue line is the extraction results of different methods.

extract the spectra of each endmember. For the more complex water spectra, the proposed algorithm accurately identifies distinct spectral signatures.

3) Jasper Ridge Dataset: The performance of different algorithms on the Jasper Ridge dataset is recorded in Tables IV and V. Similarly, the proposed algorithm demonstrates superior performance in both abundance estimation and endmember extraction. Jasper Ridge is a dataset with four endmembers, featuring more complex endmember distributions, which makes it more challenging for the algorithms. From the experimental results, unmixing models based on AEs generally achieve better results. In particular, models such as DGMSSU and the proposed method, which use endmember bundles to mitigate the effects of SV, performed exceptionally well.

The visualization results are shown in Figs. 10 and 11. According to the abundance maps, the distribution of trees and dirt in the Jasper Ridge dataset is quite complex. The proposed algorithm effectively distinguishes these two endmembers, avoiding boundary blurring and reduced purity in the abundance maps caused by misclassification. From the perspective of endmember spectral curves, dirt and road have high spectral similarity, making them prone to confusion. As shown in the figures, the introduction of endmember bundles enables the extraction of more distinctive spectra, preventing interference from lowering unmixing accuracy.

E. Urban Dataset

The experimental results on the urban dataset are presented in Tables VI and VII. The urban dataset is more complex, consisting of five typical endmembers with significant mixing between them. Based on the quantitative results, when the endmember distribution is complex, spectra mixing or various interferences intensify endmember variability. Among the various algorithms, those based on endmember bundles achieved better results, with the proposed algorithm showing the best overall performance. The visualization results are shown in Figs. 12 and 13. According to the abundance results, the distribution of grass and tree endmembers is complex and prone to misestimation. The proposed algorithm accurately distinguishes their boundaries and estimates the abundance. In addition, the roof endmember, which is sparsely distributed, covers a small and discontinuous area, leading to poorer performance in some algorithms. The proposed algorithm leverages the complementary features of endmembers, utilizing their heterogeneity to avoid inaccurate abundance estimation. The endmember spectral curves in Fig. 13 further validate the effectiveness of the proposed algorithm.

F. Discussion on Method Stability

In this section, we analyze the superiority of our bundle-based method compared to probabilistic generative models through experiments on the Urban dataset. As shown in Fig. 13, PGMSU and DGMSSU are two representative algorithms based on probabilistic generative models. Compared to PGMSU, DGMSSU introduces multiple discriminators and leverages endmember bundles to impose constraints on the generated endmembers, achieving more accurate results. However, according to the results for asphalt, roof, and dirt in Fig. 13, the curves extracted by PGMSU and DGMSSU exhibit significant fluctuations, which are detrimental to spectral recognition and matching. In contrast, the proposed method avoids irrelevant spikes in the extracted curves, yielding results that are closer to the reference endmembers. Consequently, the proposed bundle-based method produces outputs with clearer physical meanings and demonstrates greater stability.

G. Ablation Analysis

To verify the effectiveness of each module, we conduct ablation experiments on the proposed network using the Jasper Ridge dataset. The network includes an EOT, an HIF module, and a bundle-based endmember generator. In the ablation study, we replace the EOT module with a Swin Transformer

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TABLE VI
ABUNDANCE RMSE RESULTS ON URBAN DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA-FCLS	MiSiCNet	IMPSO	DOEBE	PGMSU	PPM	DGMSSU	Proposed
Asphalt	0.1715	0.1883	**0.1150**	0.1635	0.1635	0.1889	0.1974	0.1370
Grass	0.2715	0.3065	0.1828	**0.1249**	0.3021	0.2005	0.2027	0.1289
Tree	0.1510	0.3504	0.1258	0.1167	0.1926	0.1636	0.1751	**0.1166**
Roof	0.0908	0.0742	0.1151	0.1163	0.0741	0.0714	0.0617	**0.0579**
Dirt	0.3179	0.3173	0.1725	0.1646	0.1350	0.3473	**0.1065**	0.1225
Mean RMSE	0.2006	0.2474	0.1422	0.1372	0.1735	0.1943	0.1487	**0.1161**

TABLE VII
ENDMEMBER SAD RESULTS ON URBAN DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA	MiSiCNet	IMPSO	DOEBE	PGMSU	PPM	DGMSSU	Proposed
Asphalt	0.1416	0.0893	0.0777	0.1223	0.1679	0.1155	0.0739	**0.0729**
Grass	0.1127	**0.1080**	0.1307	0.1460	0.3131	0.3000	0.2097	0.1141
Tree	0.1103	0.6007	0.1085	0.1055	0.0976	0.1308	0.0577	**0.0571**
Roof	0.1388	0.0810	0.0822	0.1338	0.0871	0.0505	0.0727	**0.0382**
Dirt	1.1779	0.6995	0.1640	0.1183	0.1317	0.9831	0.0578	**0.0325**
Mean SAD	0.3363	0.3700	0.1081	0.1252	0.1595	0.3160	0.0944	**0.0630**

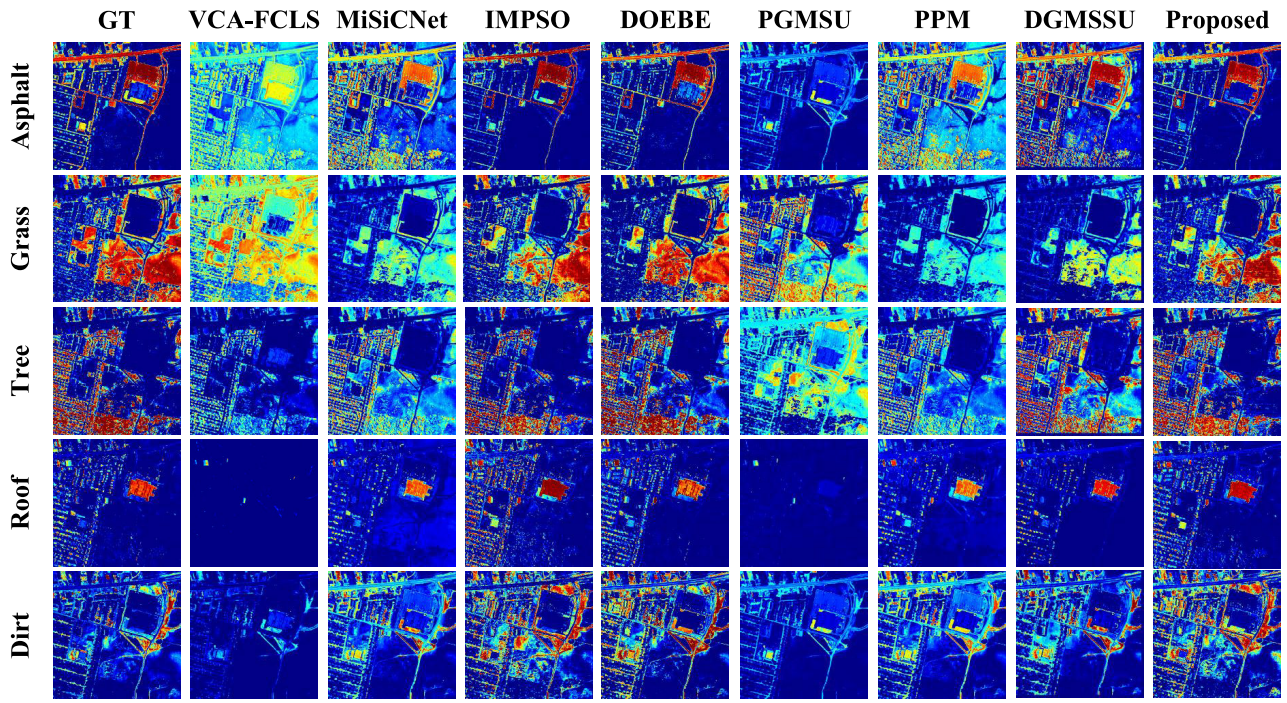


Fig. 12. Abundance map of Urban Dataset.

to perform feature extraction. For feature fusion, we employ the feature concatenation operation and a convolution-based fusion layer as alternatives. In addition, for endmember extraction, we replace the bundle-based generator with a single-layer linear layer commonly used in other unmixing AE-based networks. The results of the ablation experiments are shown in Table VIII.

Case 1 represents the baseline network composed of a Swin Transformer block and a linear decoder. In Case 2, the introduction of EOT improves both the abundance estimation and endmember extraction capabilities. When the bundle-based decoder is ablated, as seen in the comparison between Case 4 and Case 5, there is a significant performance drop, particularly in endmember extraction. This demonstrates the effectiveness of incorporating endmember bundles to account for endmember variability. Furthermore, when HIF is removed (Case 3 and Case 5), the module's inability to capture spatial complementarity between the abundance features of different endmembers leads to a substantial decline in abundance accuracy. The ablation experiments described above validate the rationality and effectiveness of each module in the network.

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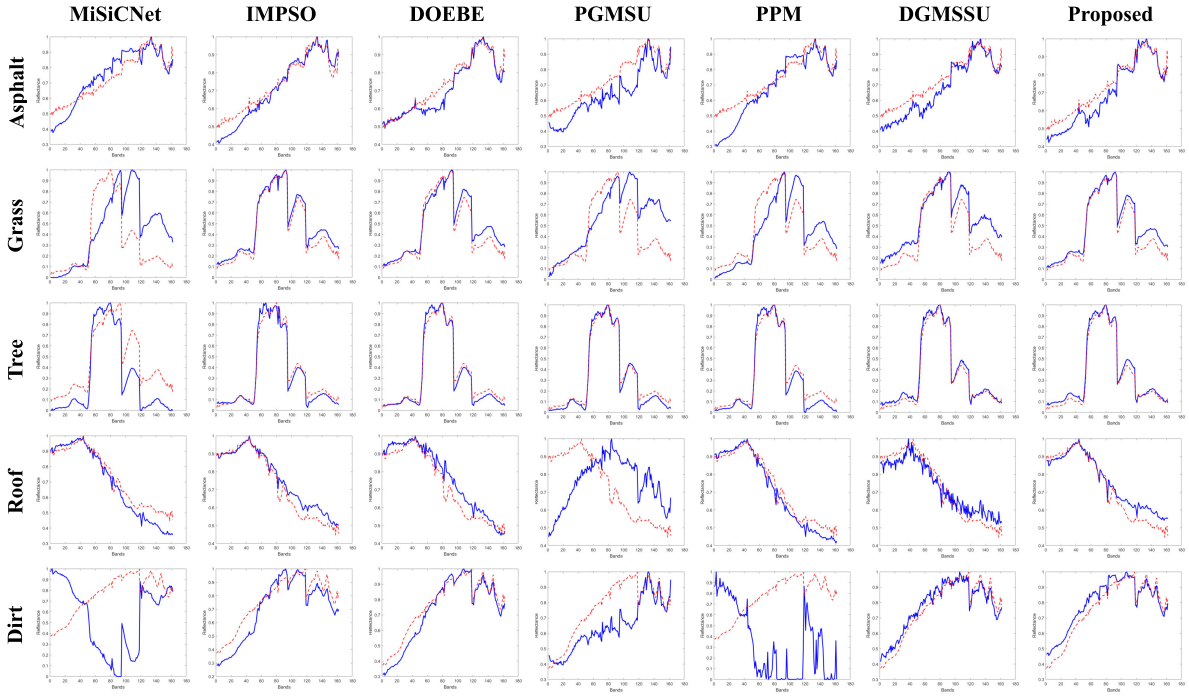


Fig. 13. Endmember curve of Urban dataset, where the red line is the true curve, and the blue line is the extraction results of different methods.

TABLE VIII
QUANTITATIVE RESULTS ON ABLATION STUDY. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Case	EOT	HIF	Bundle	aRMSE	eSAD
1	×	×	×	0.1991	0.1964
2	✓	×	×	0.1678	0.1032
3	✓	×	✓	0.0976	0.0855
4	✓	✓	×	0.0929	0.1018
5	✓	✓	✓	**0.0804**	**0.0717**

V. CONCLUSION

In this article, we propose an EOT for bundle-based unmixing. The proposed network features a bundle-based decoder, leveraging the advantages of endmember bundles to mitigate the impact of SV. To extract discriminative features for each endmember, we specifically design the EOT. This module is built on the concept of subspace projection and utilizes an LRA mechanism within each unit's subspace to extract relevant information. Furthermore, when integrating features from different endmember subspaces, we design an HIF module, incorporating prior knowledge about the complementary spatial distributions of different endmembers. This enables effective spatial and channel-wise feature interactions. This work provides a novel approach to incorporating endmember bundles into unmixing AE networks. Future work will focus on addressing the challenges in scenarios with highly mixed endmembers and developing targeted algorithms for endmember spectral extraction and separation.

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